

ECHS MATHEMATICS

IB Diploma Programme - Mathematics: Applications and Interpretation SL

1.4

Financial Applications

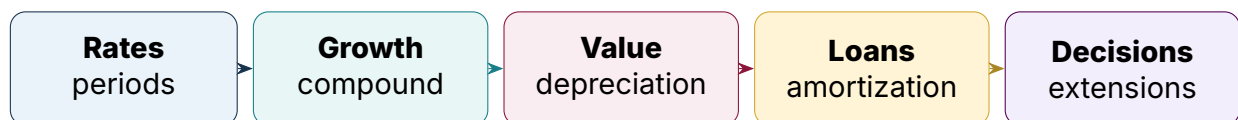
Full Notes - Compound Interest - Depreciation - Loans - Amortization - TI-84 Finance

Central idea. Financial mathematics is repeated percentage change applied to dated cash flows. A correct answer must align the **rate period, number of periods, cash-flow timing,** and a common **valuation date.**

$$A = P(1 + i)^n$$

$$V = P(1 - d)^n$$

$$B_{k+1} = B_k(1 + i) - R$$



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Version: Student Full Notes

Course scope and learning route

model → synchronize → calculate → verify → interpret

What is core in the current ECHS route?

IB SL CORE The default pathway follows the current IB Mathematics: Applications and Interpretation SL financial requirements used by this course: **compound interest and annual reducing-balance depreciation**, together with **loan repayments and amortization**. These ideas are assessed with technology used transparently.

REFERENCE EXTENSION The complete lesson also preserves effective annual rates, inflation/real value, regular deposits, annuities, withdrawal funds, outstanding-balance formulas, and sensitivity/decision analysis. These are valuable and remain available, but they are clearly labelled so they do not blur the core route.

Learning intentions

By the end of Lesson 1.4, you should be able to:

1. translate a percentage increase/decrease into a multiplicative factor;
2. distinguish simple-interest arithmetic growth from compound geometric growth;
3. match nominal annual rates, periodic rates, compounding frequency, and total periods;
4. calculate compound interest and annual depreciation, including threshold/completed-period questions;
5. set up a loan model with consistent TVM signs and timing;
6. calculate and interpret periodic repayments, interest, principal, and amortization;
7. use TI-84 Finance functions transparently and retain guard digits;
8. check reasonableness, state units/valuation dates, and evaluate assumptions.

Current-course note

The present DP Mathematics: applications and interpretation course remains the first-assessment-2021 course for students taught before the revised course begins first teaching in August 2027. The IB continues to emphasize proficiency with graphic display calculators. The notes therefore teach the mathematics first and use the GDC as a transparent tool rather than as a replacement for modelling.

1. Financial language, timelines, and percentage factors

IB SL CORE

Vocabulary you must use precisely

principal / present value	amount invested or borrowed at the valuation date
future value	amount at a stated later date
nominal annual rate	quoted yearly rate before subdivision into compounding/payment periods
periodic rate	rate applied each compounding/payment period
compounding frequency	number of compounding periods per year
growth factor	multiplier greater than 1, e.g. 1.06 for a 6% rise
retention factor	multiplier below 1, e.g. 0.88 for 12% depreciation
loan payment	regular cash flow that services interest and reduces principal
amortization	process of splitting each payment into interest and principal
outstanding balance	debt remaining immediately after a stated payment
valuation date	date to which all compared cash flows are moved

Cash-flow timeline rule

Before choosing a formula or opening TVM Solver:

1. mark time 0 and the final valuation date;
2. mark every payment/deposit date;
3. state the compounding period;
4. choose a viewpoint and assign opposite signs to opposite cash-flow directions;
5. make sure every compared value belongs to the same date.

Your Turn 1

GDC NOT NEEDED

For each statement, write the multiplicative factor.

(a) increase by 6.5% (b) decrease by 18% (c) retain 72% (d) increase by 0.8%.

WORKING / ANNOTATIONS

Additive change versus multiplicative change

A fixed money increase creates an arithmetic sequence; a fixed percentage increase creates a geometric sequence.

fixed amount: $u_{n+1} = u_n + c$

fixed percentage: $u_{n+1} = u_n(1 + r)$.

2. Simple interest as a prerequisite model

IB SL CORE

GDC NOT NEEDED

Simple interest

If principal P earns simple annual rate r for t years, the same amount Pr is added each year:

$$A = P(1 + rt), \quad I = Prt$$

The balances $P, P + Pr, P + 2Pr, \dots$ form an arithmetic sequence.

Worked Example 1

GDC NOT NEEDED

QAR 7,500 earns 3.2% simple interest for 5 years. Find the final amount and total interest.

WORKING / ANNOTATIONS

Your Turn 2

GDC NOT NEEDED

QAR 12,000 earns 4.5% simple interest for 2.5 years. Find the final amount and explain why the model is arithmetic.

WORKING / ANNOTATIONS

3. Compound interest and geometric growth

IB SL CORE

Compound-interest model

If P is compounded at periodic rate i for n periods,

$$A = P(1 + i)^n$$

If a nominal annual rate j is compounded m times per year for t years,

$$A = P \left(1 + \frac{j}{m} \right)^{mt}$$

The balances form a geometric sequence with ratio $1 + i$.

Period synchronization

Never mix annual, monthly, quarterly, or weekly quantities. The rate i and count n must refer to the **same period**. If the quote is nominal annual j with m compoundings per year, then $i = j/m$ and $n = mt$.

Worked Example 2

GDC USEFUL

QAR 14,500 is invested for 5 years at 4.8% nominal interest compounded quarterly. Find the future value.

WORKING / ANNOTATIONS

Your Turn 3

GDC USEFUL

Calculate each future value.

- (a) QAR 18,000 for 5 years at 4.6% compounded annually.
- (b) QAR 12,500 for 4 years at 5.2% nominal compounded quarterly.
- (c) QAR 9,000 for 2.5 years at 4.8% nominal compounded monthly.

WORKING / ANNOTATIONS

TI-84 Finance workflow F2

GDC SKILL

Question: QAR 10,000 for 5 years at 5% nominal compounded monthly.

1. APPS → Finance → 1:TVM Solver.
2. Enter $N = 60$, $I\% = 5$, $PV = -10000$, $PMT = 0$, $FV = 0$.
3. Set $P/Y = 12$, $C/Y = 12$. END/BEGIN does not matter because $PMT = 0$.
4. Move to FV , press ALPHA then ENTER (SOLVE).
5. Record full output first: $FV \approx 12833.58679$, then report QAR 12,833.59.

Independent check: $10000(1 + 0.05/12)^{60}$.

Common trap

In TVM Solver, enter $I\% = 5$, not 0.05, when P/Y and C/Y describe the frequency. The Finance solver expects the quoted annual percentage.

4. Completed periods, inverse questions, and thresholds

IB SL CORE

Discrete timing

Financial events often occur only at completed compounding/payment dates. If algebra or a solver gives $n = 8.31$, the model has not completed 9 periods automatically. Read the wording:

- **after how many completed periods?** test integer stages;
- **first time/value exceeds a target?** verify adjacent values;
- **continuous time permitted?** only if the context explicitly allows it.

Worked Example 3**GDC USEFUL**

QAR 20,000 grows at 6.4% nominal compounded semi-annually. Determine the least completed half-year period for the balance to exceed QAR 26,000.

WORKING / ANNOTATIONS

Your Turn 4**GDC USEFUL**

An investment of QAR 15,000 earns 5.4% nominal compounded monthly. Find the least completed month for the value to exceed QAR 18,000. Show adjacent-value verification.

WORKING / ANNOTATIONS

5. Annual reducing-balance depreciation**IB SL CORE**

Depreciation model

If an asset loses d of its value each year, it retains factor $1 - d$:

$$V_n = P(1 - d)^n$$

Depreciation is geometric decay, not repeated subtraction of the first-year loss.

Worked Example 4

GDC USEFUL

A vehicle worth QAR 84,000 depreciates by 12% annually. Find its value after 5 years.

WORKING / ANNOTATIONS

Your Turn 5

GDC USEFUL

Equipment worth QAR 48,000 depreciates by 18% annually. Find its value after 6 years.

WORKING / ANNOTATIONS

Worked Example 5

GDC USEFUL

A machine is worth QAR 72,000 and depreciates by 11% per year. Find the first year-end at which its value is below QAR 40,000.

WORKING / ANNOTATIONS

Your Turn 6**GDC USEFUL**

A vehicle worth QAR 95,000 depreciates by 14% annually.

(a) Find its value after 4 years.

(b) State why subtracting QAR 13,300 every year is wrong.

WORKING / ANNOTATIONS**Checkpoint - Compound growth and depreciation**

1. State the monthly periodic rate for 6.6% nominal compounded monthly.
2. Find QAR 24,000 after 4 years at 4.2% nominal compounded monthly.
3. A QAR 68,000 asset depreciates by 13% annually for 4 years. Find its value.
4. Explain why rounding the periodic rate before exponentiation can change an answer.
5. State one assumption of a constant-rate compound/depreciation model.

WORKING / ANNOTATIONS**6. Loan repayments: the core TVM model****IB SL CORE**

Borrower viewpoint and signs

For a borrower:

- loan received now: $PV > 0$;
- regular repayments: $PMT < 0$;
- loan fully repaid at the end: $FV = 0$;
- ordinary repayments: END mode.

The opposite global sign convention is equally valid if it is consistent.

Loan-payment mathematics

For principal PV , periodic rate i , and n equal end-of-period payments R ,

$$PV = R \frac{1 - (1 + i)^{-n}}{i}$$

so

$$R = PV \frac{i}{1 - (1 + i)^{-n}}$$

This is the present-value form of a geometric series of repayments.

Worked Example 6**GDC USEFUL**

A QAR 120,000 loan is repaid monthly over 6 years at 5.1% nominal interest compounded monthly. Find the monthly payment and total interest.

WORKING / ANNOTATIONS**TI-84 Finance workflow F6****GDC SKILL**

Question: QAR 120,000, 6 years, 5.1% nominal monthly.

1. APPS → Finance → 1:TVM Solver.

2. Enter $N = 72$, $I\% = 5.1$, $PV = 120000$, $PMT = 0$, $FV = 0$.

3. Set $P/Y = 12$, $C/Y = 12$, $PMT : END$.

4. Move to PMT and press ALPHA, ENTER (SOLVE).

5. Output: $PMT = -1938.163150 \dots$ The negative sign means money leaves the borrower.

Verification: first-month interest $= 120000(0.051/12) = 510$ QAR.

Your Turn 7

GDC USEFUL

A QAR 65,000 loan is repaid monthly for 5 years at 5.3% nominal monthly. Find the monthly payment and total interest.

WORKING / ANNOTATIONS

7. Amortization: interest versus principal

IB SL CORE

One amortization row

If B_k is the opening balance for a period, periodic rate i , and payment R :

$$I_k = iB_k, \quad P_k = R - I_k, \quad B_{k+1} = B_k - P_k = B_k(1 + i) - R.$$

Early payments contain more interest because the outstanding balance is larger.

Worked Example 7

GDC NOT NEEDED

Using the exact payment from Worked Example 6, build the first three amortization rows.

WORKING / ANNOTATIONS

Your Turn 8

GDC NOT NEEDED

A loan opens a month at QAR 40,000, the monthly rate is 0.0045, and the payment is QAR 900. Find the interest, principal repaid, and closing balance.

WORKING / ANNOTATIONS

IB communication tip

When asked to “interpret amortization”, do not only give a table. State what happens: as the balance falls, periodic interest falls and a larger part of the same payment reduces principal.

Checkpoint - Loan repayments and amortization

1. From a borrower viewpoint, state signs for PV and PMT .
2. A QAR 85,000 loan is repaid monthly over 6 years at 5.2% nominal monthly. Find the payment.
3. If an opening balance is QAR 50,000, monthly rate 0.004, and payment QAR 1,000, find one row's interest, principal, closing balance.
4. Explain why rounded PMT should not be reused to calculate a balance after many months.
5. State one financial assumption that could change a loan recommendation.

WORKING / ANNOTATIONS

8. IB-style core tasks

IB SL CORE

IB-style Task A - Compound growth and depreciation**12 marks**

A school invests QAR 24,000 for 4 years at 4.2% nominal interest compounded monthly. A separate item of equipment worth QAR 68,000 depreciates by 13% annually.

- (a) State the monthly periodic rate and the number of compounding periods. [2]
- (b) Calculate the investment value after 4 years. [2]
- (c) Calculate the equipment value after 4 years. [2]
- (d) Determine which has the larger absolute change in value. [2]
- (e) Explain why the two models use growth factors on opposite sides of 1. [2]
- (f) State one limitation of assuming constant rates. [2]

WORKING / ANNOTATIONS

IB-style Task B - Loan and amortization

14 marks

A school borrows QAR 85,000 for a vehicle. The loan is repaid monthly over 6 years at 5.2% nominal interest compounded monthly.

- State N , $I\%$, PV , FV , P/Y , C/Y , and payment timing from the borrower viewpoint. [3]
- Calculate the monthly payment. [2]
- Calculate the first month's interest and principal repaid. [3]
- Find the closing balance after the first payment. [2]
- Calculate the total amount repaid over 72 payments and hence total interest. [2]
- State one assumption affecting the reliability of the calculation. [2]

WORKING / ANNOTATIONS

9. Core misconception clinic and mastery

IB SL CORE

Misconception clinic

"5% up then 5% down cancels." False: $1.05(0.95) = 0.9975$, so the result is 0.25% below the start.

"Depreciation subtracts the same QAR amount yearly." False: the same percentage is applied to a changing balance.

"TVM $I\%$ should be divided by 12 before entry." Not when using TVM Solver with $P/Y = C/Y = 12$; enter the quoted nominal annual percent.

"A negative PMT is an error." No: it represents cash-flow direction relative to the chosen viewpoint.

"Round PMT to cents immediately." Report cents, but keep full solver precision for later balance/amortization calculations.

"A calculator output is a complete answer." No: state the model, settings, units, timing, and interpretation.

Checkpoint - Core mastery

1. QAR 30,000 earns 4.75% nominal compounded quarterly for 7 years. Find the final value.
2. An asset worth QAR 52,000 depreciates by 9% annually for 6 years. Find its value.
3. A QAR 90,000 loan is repaid monthly over 60 months at 4.9% nominal compounded monthly. Find the payment.
4. For that loan, calculate the first month's interest and principal repaid.
5. Write an IB-quality conclusion for part 3 including sign interpretation and timing.

WORKING / ANNOTATIONS

Reference-supported extensions

REFERENCE EXTENSION

How to use this section

These topics are preserved in the complete ECHS lesson because they strengthen modelling and financial reasoning. They are clearly marked **REFERENCE EXTENSION** so students can distinguish enrichment from the current default core route. They are especially useful for broader applications, projects, and richer GDC fluency.

10. Effective annual rate and advertised-rate comparison

REFERENCE EXTENSION

GDC SKILL

Effective annual rate

For nominal annual rate j compounded m times per year,

$$r_{\text{eff}} = \left(1 + \frac{j}{m}\right)^m - 1$$

This gives the one-year percentage actually produced by compounding.

Worked Example 8

GDC USEFUL

Find the effective annual rate for 4.8% nominal interest compounded monthly.

WORKING / ANNOTATIONS

TI-84 Finance workflow F1

GDC SKILL

APPS → Finance → `eff(`. Enter `eff(4.8,12)` and press ENTER. The Finance function uses percentage form, so enter 4.8, not 0.048. Output $\approx 4.907020753\%$. Verify with $(1 + 0.048/12)^{12} - 1$.

Your Turn 9

GDC USEFUL

Compare 6.1% nominal compounded monthly with 6.2% compounded annually. Which gives the larger one-year return?

WORKING / ANNOTATIONS

11. Inflation, real value, and purchasing power

REFERENCE EXTENSION

Nominal versus real value

If inflation is f per year for n years,

$$\text{real value} = \frac{\text{nominal future value}}{(1 + f)^n}$$

For an exact effective real return,

$$1 + r_{\text{real}} = \frac{1 + r_{\text{nominal}}}{1 + f}$$

Worked Example 9

GDC USEFUL

An investment grows to QAR 28,382.15 after 4 years while inflation averages 3.4% annually. Find its real value in base-year QAR.

WORKING / ANNOTATIONS

Your Turn 10

GDC USEFUL

A future amount is QAR 30,000 after 5 years of 2.5% annual inflation. Find its real value.

WORKING / ANNOTATIONS

12. Regular deposits and ordinary annuities

REFERENCE EXTENSION

Future value of end-of-period deposits

Equal end-of-period deposits R accumulate as a geometric series:

$$FV = R + R(1+i) + \cdots + R(1+i)^{n-1}$$

so

$$FV = R \frac{(1+i)^n - 1}{i}$$

Worked Example 10

GDC USEFUL

Deposit QAR 600 at each month-end for 8 years at 5.4% nominal compounded monthly.

WORKING / ANNOTATIONS

TI-84 Finance workflow F3

GDC SKILL

TVM Solver: $N = 96$, $I\% = 5.4$, $PV = 0$, $PMT = -600$, $FV = 0$, $P/Y = C/Y = 12$, $PMT: END$. Solve FV . Output $FV \approx 71,845.74787$. Compare with total contributions $600(96)$.

Your Turn 11

GDC USEFUL

QAR 300 is deposited monthly for 10 years at 4.8% nominal compounded monthly. Find future value, contributions, and interest earned.

WORKING / ANNOTATIONS

13. Annuity due: beginning-of-period deposits

REFERENCE EXTENSION

Timing shift

Every beginning-of-period deposit earns one extra period:

$$FV_{due} = R \frac{(1+i)^n - 1}{i} (1+i)$$

The number of deposits remains n ; only timing changes.

Worked Example 11

GDC USEFUL

Deposit QAR 500 at the beginning of each month for 5 years at 6% nominal monthly.

WORKING / ANNOTATIONS

TI-84 Finance workflow F4

GDC SKILL

TVM Solver: $N = 60$, $I\% = 6$, $PV = 0$, $PMT = -500$, $FV = 0$, $P/Y = C/Y = 12$, then select PMT: BEGIN and solve FV . Return the calculator to END mode after the problem unless the next problem is also beginning-of-period.

Your Turn 12

GDC USEFUL

QAR 1,500 is deposited at the beginning of each year for 25 years at 2.49% annually. Find the future value.

WORKING / ANNOTATIONS

14. Mixed savings and target balances

REFERENCE EXTENSION

Lump sum plus regular deposits

For an initial amount P plus end-of-period deposits R ,

$$FV = P(1 + i)^n + R \frac{(1 + i)^n - 1}{i}$$

Treat the two cash-flow components separately, then combine them at the same valuation date.

Worked Example 12

GDC USEFUL

An account begins with QAR 4,800. QAR 180 is deposited at each month-end for 3 years at 4.8% nominal monthly.

WORKING / ANNOTATIONS

Your Turn 13

GDC USEFUL

QAR 10,500 is invested initially and QAR 350 is added monthly for 4 years at 5.7% nominal monthly. Find the final balance.

WORKING / ANNOTATIONS

Least number of deposits

For a target T , solve

$$R \frac{(1+i)^n - 1}{i} > T,$$

then choose and verify the first admissible whole number of deposits. Always check FV_{n-1} and FV_n .

15. Withdrawal funds

REFERENCE EXTENSION

Present value of equal withdrawals

For n end-of-period withdrawals R supported by a fund today,

$$PV = R \frac{1 - (1+i)^{-n}}{i}$$

Equivalently,

$$R = PV \frac{i}{1 - (1+i)^{-n}}$$

Worked Example 13

GDC USEFUL

QAR 500,000 earns 5.25% nominal compounded monthly. Find the month-end withdrawal that exhausts the fund after 25 years.

WORKING / ANNOTATIONS

TI-84 Finance workflow F5

GDC SKILL

From the retiree viewpoint use $N = 300$, $I\% = 5.25$, $PV = -500000$, $PMT = 0$, $FV = 0$, $P/Y = C/Y = 12$, END. Solve PMT . The output magnitude is 2996.238576. Opposite signs between PV and PMT are essential.

Your Turn 14

GDC USEFUL

A €350,000 fund earns 5.8% nominal compounded monthly. Find the monthly withdrawal that lasts 20 years.

WORKING / ANNOTATIONS

16. Outstanding balance and cumulative loan evidence

REFERENCE EXTENSION

GDC SKILL

Retrospective balance

After k equal payments,

$$B_k = PV(1+i)^k - R \frac{(1+i)^k - 1}{i}$$

A prospective method values the remaining payments instead. Both must agree when timing is consistent.

Worked Example 14**GDC USEFUL**

A QAR 90,000 loan at 4.9% nominal monthly is repaid over 60 months. Find the balance immediately after payment 24.

WORKING / ANNOTATIONS**TI-84 Finance workflow F7****GDC SKILL**

After setting the loan in TVM Solver and solving PMT:

- APPS $\rightarrow$ Finance $\rightarrow$ bal(24) gives the balance after payment 24;
- $\Sigma\text{Int}(1, 24)$ gives cumulative interest;
- $\Sigma\text{Prn}(1, 24)$ gives cumulative principal.

For this loan: $B_{24} \approx 56,615.98$, cumulative interest magnitude $\approx 7,278.96$, cumulative principal $\approx 33,384.02$ QAR. Check $90,000 - 33,384.02 \approx 56,615.98$.

Your Turn 15**GDC USEFUL**

A QAR 60,000 loan at 5.2% nominal monthly is repaid over 60 months. Find the balance after 24 payments.

WORKING / ANNOTATIONS**17. Financial decisions, assumptions, and sensitivity****REFERENCE EXTENSION**

A defensible financial recommendation

A strong recommendation should state:

1. what is being compared and at what common valuation date;
2. the rate convention and timing assumptions;
3. any fees, taxes, inflation, or penalties included/excluded;
4. the numerical result with sensible currency rounding;
5. a sensitivity statement: would a plausible change in rate or payment reverse the decision?

Your Turn 16

GDC NOT NEEDED

A recommendation changes when the assumed interest rate moves from 4.7% to 4.8%. What should you conclude about robustness?

WORKING / ANNOTATIONS

Model limitations to mention in IB responses

Real financial outcomes can differ because of variable interest/inflation, fees, taxes, missed payments, changing exchange rates, penalties, asset-market shocks, and institutional rules. A good model is useful *because* its assumptions are explicit, not because it predicts perfectly.

Checkpoint - Extension mastery

1. Find the EAR for 6.0% nominal compounded monthly.
2. Find the future value of QAR 1,200 deposited quarterly for 6 years at 5.2% nominal quarterly.
3. Find the present fund required for QAR 6,000 monthly withdrawals for 25 years at 5.25% nominal monthly.
4. Explain the difference between END and BEGIN annuity timing.
5. State how to verify a TI-84 outstanding-balance result independently.

WORKING / ANNOTATIONS**18. Mixed IB-style challenge tasks**

core + extension clearly identified

Challenge Task C - Advertised rate and savings**REFERENCE EXTENSION 12 marks**

A savings provider quotes 5.0% nominal interest compounded monthly. A student deposits QAR 900 at each month-end for 6 years.

- (a) Calculate the effective annual rate. [2]
- (b) Find the value immediately after the final deposit. [3]
- (c) Find total contributions and interest earned. [2]
- (d) Find the value if deposits were made at the beginning of each month. [2]
- (e) State two reasons the actual outcome could differ from the model. [2]
- (f) Explain why comparing this balance with another plan's balance one year later would be invalid without adjustment. [1]

WORKING / ANNOTATIONS

Challenge Task D - Loan decision**IB SL CORE 13 marks**

Two lenders offer a QAR 100,000 loan at 5.6% nominal compounded monthly. Lender A requires repayment over 5 years; Lender B over 8 years.

- Find the monthly payment for each option. [4]
- Find total interest for each option. [3]
- State the main trade-off between the two plans. [2]
- A family can afford at most QAR 1,600 per month. Which option is feasible? [2]
- State one non-mathematical factor that should also be considered. [2]

WORKING / ANNOTATIONS**19. Formula and TI-84 reference**

ready for revision

Core formula reference

simple interest: $A = P(1 + rt)$

compound interest: $A = P(1 + i)^n = P \left(1 + \frac{j}{m} \right)^{mt}$

annual depreciation: $V = P(1 - d)^n$

loan payment: $R = PV \frac{i}{1 - (1 + i)^{-n}}$

amortization interest: $I_k = iB_k$

principal repaid: $P_k = R - I_k$

new balance: $B_{k+1} = B_k(1 + i) - R$

Extension formula reference

effective annual rate: $r_{eff} = \left(1 + \frac{j}{m}\right)^m - 1$

real value: $V_{real} = \frac{V_{nominal}}{(1 + f)^n}$

ordinary annuity FV: $FV = R \frac{(1 + i)^n - 1}{i}$

annuity due FV: $FV_{due} = FV_{ord}(1 + i)$

annuity PV: $PV = R \frac{1 - (1 + i)^{-n}}{i}$

retrospective balance: $B_k = PV(1 + i)^k - R \frac{(1 + i)^k - 1}{i}$

TI-84 Finance quick reference

GDC SKILL

TVM Solver

APPS → Finance → 1:TVM Solver; set $N, I\%, PV, PMT, FV, P/Y, C/Y$, END/BEGIN; move to unknown; ALPHA ENTER (SOLVE).

eff(

nominal percentage and compounding frequency, e.g. eff(4.8,12).

bal(

outstanding balance after a stated number of payments, once the TVM model is stored.

ΣInt(

cumulative interest over a payment range.

ΣPrn(

cumulative principal over a payment range.

Sign convention

opposite cash-flow directions must have opposite signs.

Precision

keep full calculator values internally; round only the reported final money amount unless instructed otherwise.

20. Exit ticket and reflection

independent mastery evidence

Exit ticket

1. Explain in one sentence why compound interest is geometric.
2. Write the model for 9% annual depreciation.
3. State the correct TVM signs for a borrower who receives a loan and makes repayments.
4. Explain the difference between interest and principal in one amortization payment.
5. Give one example of a GDC check that is stronger than merely repeating the same calculator command.
6. Mark each topic as **IB SL CORE** or **REFERENCE EXTENSION**: compound interest; annual depreciation; loan repayments; amortization; effective annual rate; annuity due; withdrawal fund; sensitivity analysis.

WORKING / ANNOTATIONS

Final IB communication checklist

Before submitting a financial answer, ask:

- Did I state the rate period?
- Did I round currency sensibly?
- Did I synchronize i and n ?
- Did I give units and a valuation date?
- Did I use a consistent sign convention?
- Did I interpret the sign/output?
- Did I identify END or BEGIN timing?
- Did I verify with another representation?
- Did I retain guard digits?
- Did I state assumptions/limitations where relevant?

Lesson 1.4 complete

Financial mathematics is not a list of buttons. It is a model of rates, periods, cash flows, timing, verification, and interpretation.